

FSCT8561-Lab0-AlvinSin

Reflection Question 1

1. What is a socket?

A socket is an endpoint of communication defined by IP address and port number, it is an interface between the application and the network allowing data to be passed to the socket once it is setup on the application for network transmission

2. What does `AF\_INET` mean?

AF_INET is an address family that is used to designate the type of addresses that the socket can communicate with, in this case its on IPv4. When creating a socket the address family has to be specified and can only use the addresses of that type of the socket

3. What does `SOCK\_STREAM` mean?

SOCK_STREAM means it is a TCP socket

Part 5

```
CT-8561/labs/Lab0$ python3 server.py
Server is waiting for a connection...
Connected by: ('127.0.0.1', 58530)
Client says: Hello Server
```

```
CT-8561/labs/Lab0$ python3 client.py
Server replied: Message received
```

Part 6

```
CT-8561/labs/Lab0$ python3 server.py
Server is waiting for a connection...
Connected by: ('127.0.0.1', 34326)
Client says: Hello Server, my name is Alvin
```

```
CT-8561/labs/Lab0$ python3 client.py
Server replied: Server received: Hello Server, my name is Alvin
```

Reflection Question 2

1. Which host did the client connect to?

Cloudflare

2. Which port was used?

Port80

3. Why is port 80 relevant?

Using port 80 to connect to the web server using HTTP, using other ports would lead to different service

4. What did the server send back?

HTML of the webpage

```
22f
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon"
href="data:,"><meta name="viewport" content="width=device-width, initial-scale=1"><
style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-s
erif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head>
<body><div><h1>Example Domain</h1><p>This domain is for use in documentation exampl
es without needing permission. Avoid use in operations.</p><p><a href="https://iana
.org/domains/example">Learn more</a></p></div></body></html>
```

Part 8

```
import socketio
from aiohttp import web

# Create the Socket.IO server
sio = socketio.AsyncServer(
    async_mode="aiohttp",
    cors_allowed_origins="*"
)

# App start
app = web.Application()
sio.attach(app)

# Send message to client when connected
@sio.event
async def connect(sid, environ):
    print(f"Client connected: {sid}")
    await sio.emit("message", "Welcome!", to=sid)

# Logs when client disconnects
@sio.event
async def disconnect(sid):
    print(f"Client disconnected: {sid}")
```

```
# Handles message from client
@sio.event
async def message(sid, data):
    print(f"Received from {sid}: {data}")

    # Send a response back to that client
    await sio.emit(
        "message",
        f"Server received: {data}",
        to=sid
    )

if __name__ == "__main__":
    web.run_app(app, host="0.0.0.0", port=5000)
```

```
# Client
import socketio
sio = socketio.Client()

@sio.event
def connect():
    print("Connected to server")

    # Send a message to the server
    sio.emit("message", "Hello from the client!")

@sio.event
def disconnect():
    print("Disconnected from server")

@sio.event
def message(data):
    print("Server says:", data)

sio.connect("http://localhost:5000")
sio.wait()
```

```
CT-8561/labs/Lab0$ python3 server.py
===== Running on http://0.0.0.0:5000 =====
(Press CTRL+C to quit)
Client connected: LC6tWXqr3pF2Se0dAAAB
Received from LC6tWXqr3pF2Se0dAAAB: Hello from the client!
Client disconnected: LC6tWXqr3pF2Se0dAAAB
█
```

```
CT-8561/labs/Lab0$ python3 client.py
websocket-client package not installed, only polling transport is available
Server says: Welcome!
Connected to server
Server says: Server received: Hello from the client!
^CDisconnected from server
```

1. What could happen if a server accepts connections from anyone?

User who are unauthorized can connect and send malicious request and access function that they should not be using.

2. Is the Lab 0 message encrypted?

No, as it is using HTTP, the message is not protected by TLS encryption, message can be seen in plain text.

3. Could network traffic be observed by another system?

Yes, as this is on an unencrypted network connection, potential attacker can capture and inspect the traffic

4. Why should servers validate input?

Validating input to ensure received data is in expected formats and limitations, help prevent formats that are not support, or malicious intent request or security vulnerability

5. Why might authentication be useful?

Authentication will verify who the client is, this will restrict access to users that are only authorized and can apply permissions based on the user's identify

6. Suggest two ways to make this application more secure.

Using TLS encryption through HTTPS and adding authentication to protect the CIA triad

Reflection

This lab has taught me how 2 device or programs can communicate with each other over a network with a client and a server, learning how sockets use IP and port to connect with each other. This lab is fairly simple but it give us better understanding of what is happening behind the scenes when applications communicate over a network and provided a foundation for learning more advanced cybersec concepts.